ICT2152 · Object Oriented Analysis and Design

Foundations: Requirements and Domain Models

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

Analysis identifies stakeholders, goals, constraints, and observable behavior before selecting implementation details. A domain model captures important concepts and relationships. A use case describes a goal-oriented interaction and its successful and alternate paths.

2. Example and reasoning

For attendance capture, identify the technical officer, course offering, session, enrolled student, and attendance record. State authorization and validation conditions as part of the use case, not as assumptions.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “foundations: requirements and domain models” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.